Individual Contribution Statements

ICT 5143 Capstone, Problem 8

Sumedh Navuda and Aldino

25 October 2026

The work was divided by pipeline stage. Sumedh Navuda led the question, literature, data, predictive Bayesian network, exact inference and the Texas test of H1. Aldino led the Bayesian audit models, the simulations, threshold selection, uncertainty analysis and the improved audit procedure (H2). Each student reviewed the other's stages and can explain the whole pipeline. Section numbers refer to `Problem8_Intersectional_Fairness.ipynb`.

Contribution table

Work item	Lead	Reviewer	Artefact	What the reviewer verified
Research question, preregistration (ϵ, capacity, utility budget, confirmatory family)	Sumedh Navuda	Aldino	notebook top cells, `docs/proposal.md`	confirmatory family matches §11 Holm family
Literature review (19 papers), comparison table, narrowed claim, H1	Sumedh Navuda	Aldino	`writeup/literature_review.tex`	closest-work rows against the struct and contamination audits
Data acquisition, label reversal, data card, household split	Sumedh Navuda	Aldino	§1–§2	no SERIALNO in two splits; label is exactly $1-\text{PUBCOV}_{=1}$
Predictive BN, VE, clique-tree check, LR/GBT baselines	Sumedh Navuda	Aldino	§3–§4.1	VE vs clique tree 3.3×10^{-16}; CI check
Experiment 1 (single vs intersectional)	Sumedh Navuda	Aldino	§5, `exp1_*.csv`	21 of 35 test groups with ≥ 30 positives
Audit models: EB, full Bayes, struct, contamination prior; verification	Aldino	Sumedh Navuda	§6, §6.1	beta-binomial posterior mean by hand; quadrature and gradient checks
Experiments 3 and 4, calibration, semi-synthetic worlds, robustness	Aldino	Sumedh Navuda	§7–§9.2, §12	frozen §9 rule reused unchanged in §9.2
Experiments 2 and 5, bootstrap, Holm, survey weights	Aldino	Sumedh Navuda	§10, §11, §13	Holm p recomputed (0.368 $\rightarrow$ 0.736)
Texas replication and H1 test	Sumedh Navuda	Aldino	§14, `h1_texas_test.csv`	decision rule fixed before the run; 27/27 count

Work item	Lead	Reviewer	Artefact	What the reviewer verified
Improved audit (design-matched calibration, held-out certification), H2	Aldino	Sumedh Navuda	§15, `v2_*.csv`	J recomputed from `v2_eval_*.csv`; paired CI
Manuscript, report, slides	both	both	`writeup/`, `report/`, `slides/`	every number traced to `results/`
Model card, assumptions register	Aldino	Sumedh Navuda	`writeup/`	statuses match notebook evidence
README, repository, reproducibility	Sumedh Navuda	Aldino	`README.md`, `docs/`	setup commands and manifest

Sumedh Navuda ([roll no.])

Research design Research question, preregistration table (30% capacity, $\epsilon = 0.10$, recall budget 0.02, confirmatory family) and the falsifiable contribution H1 with its Texas decision rule (§14.1).

Literature review 19 papers, the landscape comparison table and gap verification; narrowed the claim to an evaluation after finding that hierarchical intersectional estimation (Foulds et al.) and contamination priors (Berger and Berliner) already exist.

Data and predictive BN Folktables acquisition and checks, data card, household-disjoint splits, discretisation, the factor and variable-elimination code, the BN, the inference verification against pgmpy and the LR/GBT baselines.

Experiments Experiment 1; Texas replication and the H1 test (27 of 27 pooled claims failed on test).

Writing Manuscript introduction, data and model sections; literature review; report sections 1–2 and reproducibility; slides 1–5; README and repository.

Technical decision Splitting by household (SERIALNO) rather than by person, so that people who share circumstances never appear on both sides of a split.

Difficulty or failure The earlier draft claim (the contamination prior halves false certification) failed: 0.503 against the required 0.490. We kept the failure on record, narrowed the claim, and tested H1 out of sample.

Cross-check of Aldino's work Recomputed the beta-binomial posterior mean by hand ($k = 2$, $n = 10$, $\mu = 0.4$, $\kappa = 90$ gives 0.38), recomputed the Holm adjustment, and recomputed J and its paired interval from `results/v2_eval_*.csv`.

Tools and assistance Python (numpy, scipy, pandas, pgmpy, folktables), Jupyter, LaTeX. An AI coding assistant (Claude Code) helped draft code, LaTeX and documentation; I checked every reported number against the saved notebook outputs in `results/`.

Signature and date _____

Aldino ([roll no.])

Audit models Beta-binomial empirical Bayes, the full-Bayes grid posterior, the attribute-structured audit (`struct`: marginal likelihood by adaptive quadrature with an exact gradient) and the contamination prior; their numerical and coverage checks (§6.1).

Experiments Experiment 3 on real counts; Experiment 4 (synthetic and semi-synthetic worlds) and the preregistered calibration (§9); Experiments 2 and 5 (threshold family, minority thinning, selection); the household bootstrap with Holm correction; survey-weighted estimates; robustness grids.

Improved procedure Diagnosis of the false alarms (90% come from groups with fewer than 10 positives), design-matched calibration, held-out certification and the H2a–c tests (§15).

Writing Manuscript audit, experiment and improvement sections; model card; assumptions register; report sections 3–5; slides 6–10.

Technical decision Ranking audit procedures by $J = \mathrm{ACC} - \mathrm{FTS}$ fixed before calibration, so that a procedure cannot win by never certifying (score 0) or always certifying (score 0).

Difficulty or failure The calibration chosen on invented group sizes did not transfer to the real design: fair worlds were accepted 83% of the time on the invented design and only 32–33% on the California design. This motivated calibrating on the target design.

Cross-check of Sumedh Navuda's work
Checked that no household appears in two splits, that the label is the exact reversal of the Folktables label, that VE, pgmpy VE and the clique tree agree with enumeration, and the group denominators of Experiment 1.

Tools and assistance Python (numpy, scipy, pandas), Jupyter, LaTeX. An AI coding assistant (Claude Code) helped draft code, LaTeX and documentation; I checked every reported number against the saved notebook outputs in `results/`.

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